

BIOMECHANICS (EPHE 341)

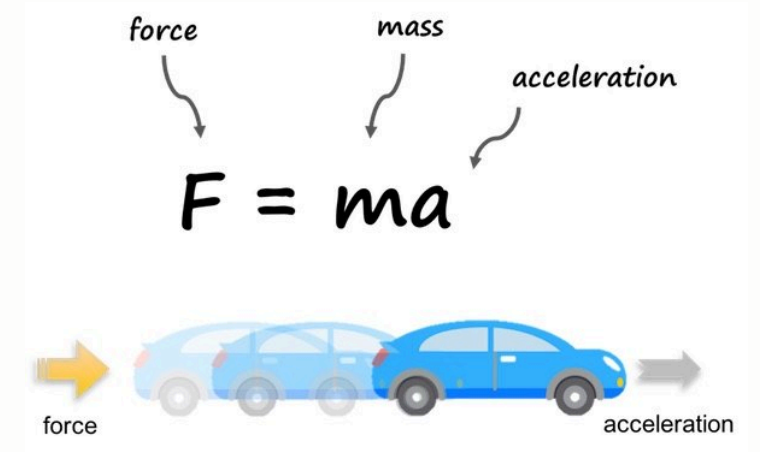
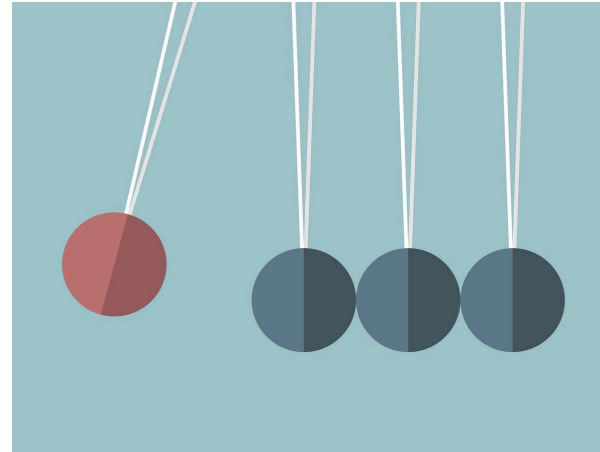
Linear Kinetics

Optional reading: McGinnis Chpt 3

Objectives

To learn/review;

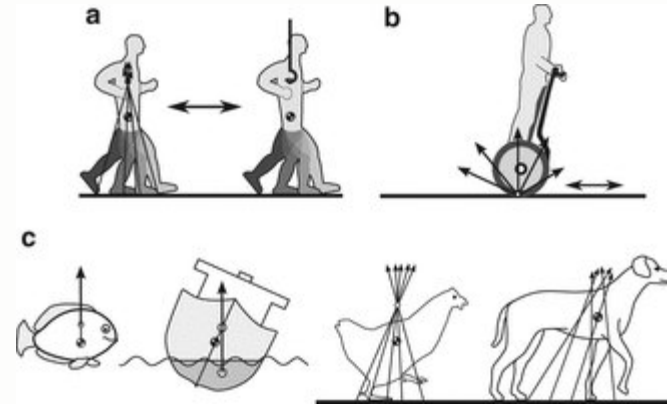
- Newton's 3 laws
- Kinetics equations and relationships
- Force
- Momentum
- Impulse
- Coefficient of restitution



1st Law – Inertia · 2nd Law – Acceleration · 3rd Law – Action/Reaction

Kinetics:

The concepts of mass, force, and energy as they affect motion.



Newton's 1st Law : Law of Inertia

An object at rest tends to stay at rest and an object in uniform motion tends to stay in uniform motion unless acted upon by a **net external force**.

- uniform = unchanging = consistent
- Net = total after all forces are added together

What is Inertia?

- derived from the latin for "idleness" or "laziness"
- resistance to a change in state of motion
- Mass quantifies inertia (measure of inertia)



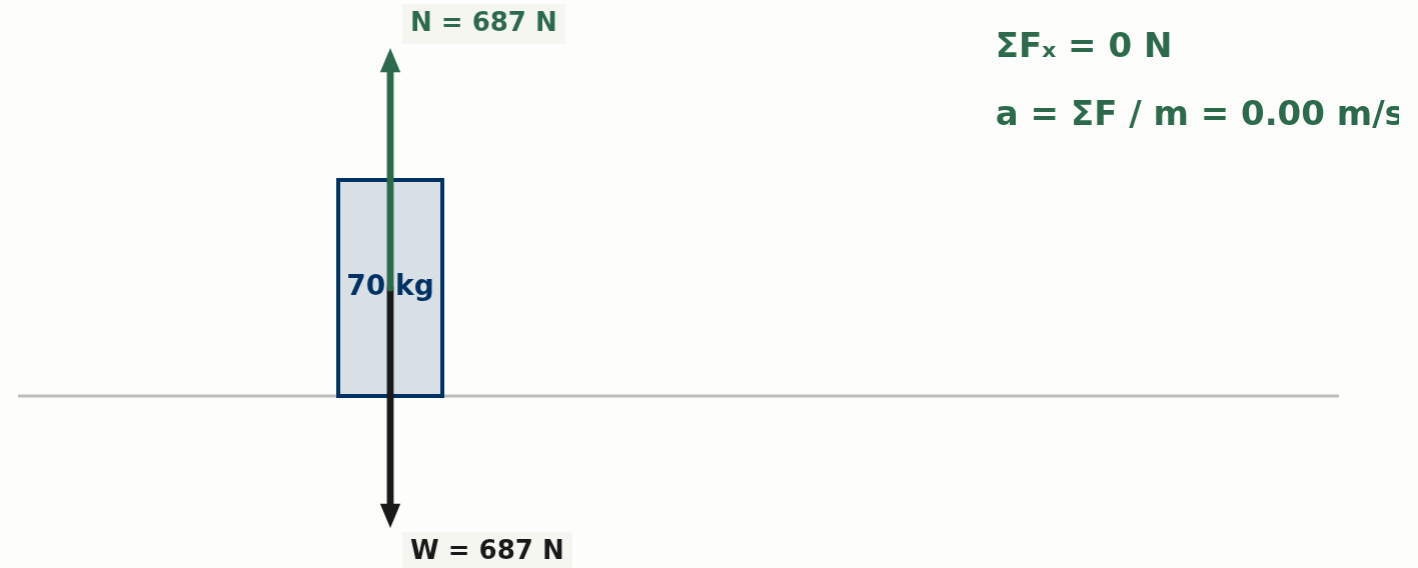
Static equilibrium

An object is at rest when the net forces acting on it sum to zero.

- There is no motion
- Gravity is acting downward
- The ground contact force is acting up
- The net vertical force is zero

$$\Sigma F_z = 0 \quad \Sigma F_x = 0 \quad \Sigma F_y = 0$$

This individual is in static equilibrium. **Is there such a thing as dynamic equilibrium?**



static equilibrium — at rest, and staying that way

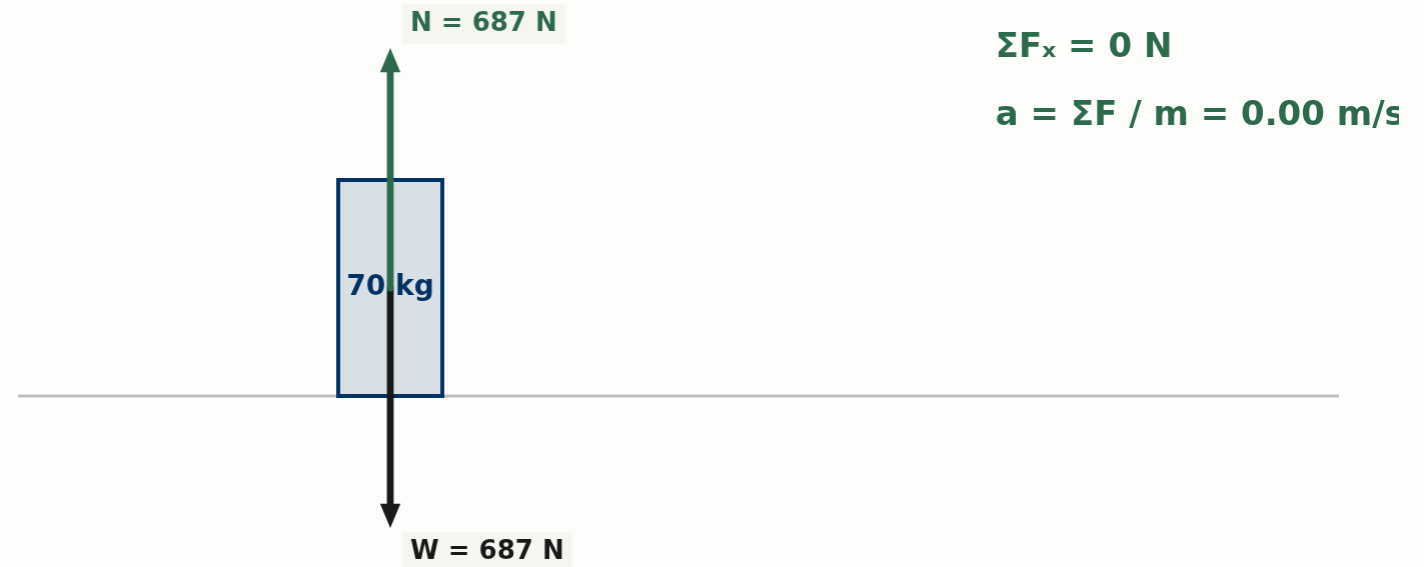
The force diagram for a body standing still and a body gliding at constant velocity is the **same diagram**. Nothing in the forces distinguishes them, which is exactly what the first law says.

Body mass 70 kg
Net horizontal force 0 N

Dynamic equilibrium

When is an object in uniform motion? This is **Dynamic Equilibrium**.

- Uniform motion means constant velocity.
- An object will maintain its velocity unless accelerated or decelerated.
- Net forces are equal to zero but an object is in motion



dynamic equilibrium — moving, and staying that way

The force diagram for a body standing still and a body gliding at constant velocity is the **same diagram**. Nothing in the forces distinguishes them, which is exactly what the first law says.

Body mass 70 kg
Net horizontal force 0 N

Momentum

The momentum of an object can be thought of as the tendency for an object to resist change in motion and continue to move in its direction of travel.



The football player's momentum will resist change in the state of motion.
He will be difficult to decelerate



The cyclist's momentum will resist change in the state of motion. He will
keep moving forward

Momentum

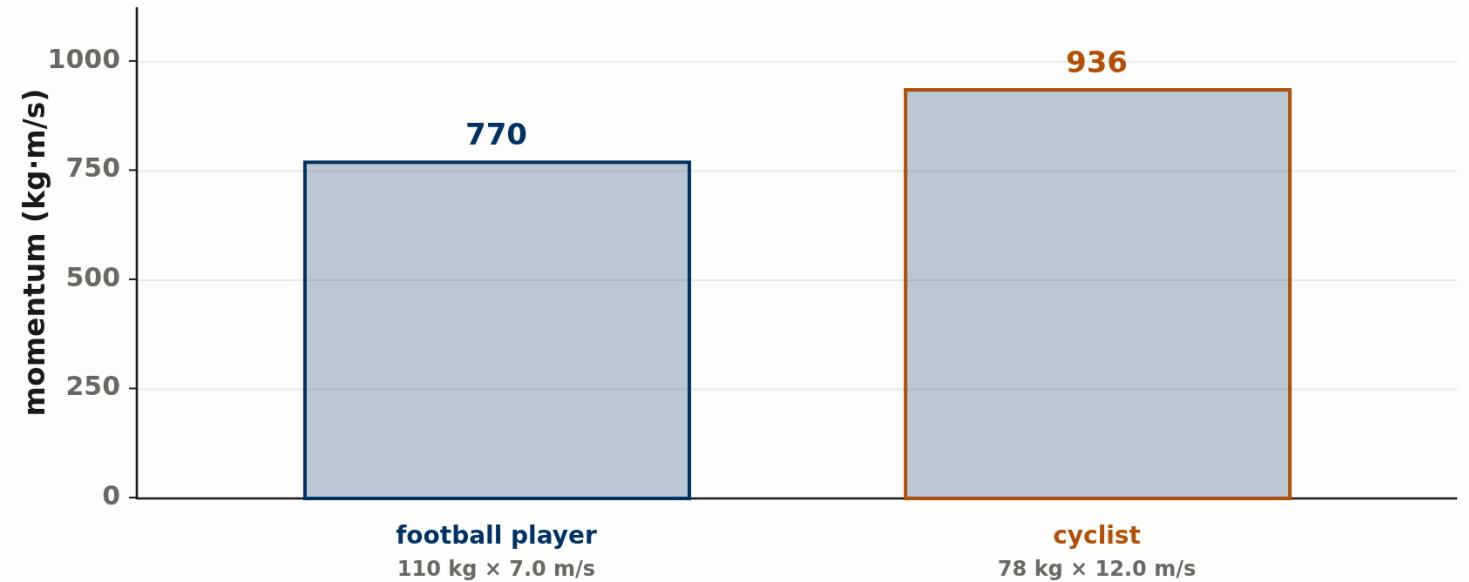
Momentum is related to inertia but not the same. Inertia is an object's tendency to resist change in the state of motion (at motion and rest). However during motion an object has momentum.

That is, an object resists its change in state of motion and this resistance is related to its **mass** (inertia) and **velocity**.

$$P = mv$$

where P is the object's linear momentum, m is the object's mass and v is the object's velocity. SI units: kg·m/s

Remember that velocity is a vector quantity, so momentum will have a magnitude and direction.



$$P = mv \cdot 770 \text{ against } 936 \text{ kg} \cdot \text{m/s}$$

Momentum is mass **and** velocity. Make the lighter one fast enough and it carries as much as the heavier one — try it. Velocity is a vector, so momentum has a direction too.

Player speed 7.0 m/s
Cyclist speed 12.0 m/s

Player mass 110 kg

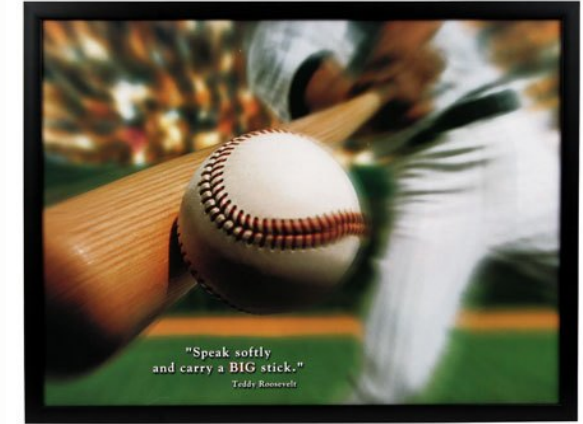
Cyclist mass 78 kg

Momentum

Understanding Momentum is really important when dealing with things that collide.

- Baseball
- Soccer
- Football players

The outcome of collisions during these sports can be explained by quantifying the momentum of the system.



Law of Conservation of Momentum

In the absence of external forces, the total momentum of a given system is conserved.

$$P_{\text{before}} = P_{\text{after}} \quad \text{or} \quad (mv)_{\text{before}} = (mv)_{\text{after}}$$

The law of conservation of momentum helps us understand collisions — the sum of the momentum before the collision must equal the sum of the momentum after the collision.

Collisions range between elastic and inelastic collisions:

- **Elastic collisions** — objects rebound off of each other
- **Inelastic collisions** — objects stick together and usually become distorted and generate heat



Coefficient of restitution

Most collisions are neither perfectly elastic nor perfectly inelastic but somewhere in between.

The **coefficient of restitution** is a means of quantifying how elastic the collisions are.



- The coefficient of restitution is defined as the absolute value of the ratio of the **velocity of separation** to the **velocity of approach**.
- The velocity of approach is the difference between the velocities of the two colliding objects just before the collision (ie how fast they are moving toward each other).
- The velocity of separation is the difference between the velocities of the two colliding objects just after the collision (ie how fast they are moving away from each other).

e = velocity of separation / velocity of approach

$$= (V_{Bf} - V_{Af}) / (V_{Ai} - V_{Bi})$$

Where V_{Bf} = final velocity of object B, V_{Af} = final velocity of object A, V_{Ai} = initial velocity of object A, V_{Bi} = initial velocity of object B.

- The coefficient of restitution is a ratio and therefore has no units.
- For perfectly elastic collision the COR is **1**
- For perfectly inelastic collisions the COR is **0**

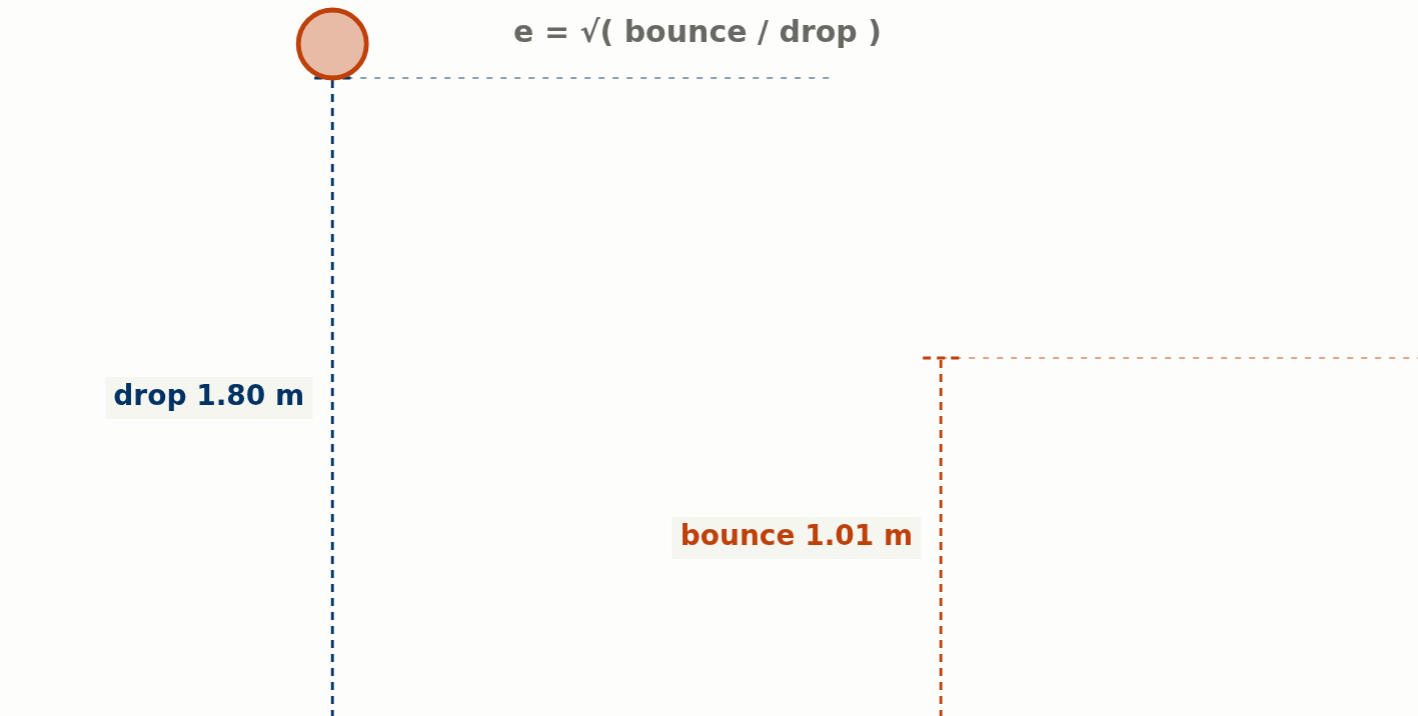
Most rules for ball sports directly or indirectly specify what the coefficient of restitution must be for the ball on the specific playing surface or implement.

$$\text{coefficient of restitution} = e = \sqrt{\frac{\text{bounce height}}{\text{drop height}}}$$

Coefficient of restitution

The coefficient of restitution is affected by the nature of both objects. For ball sports the coefficient of restitution can be figured out using a simple equation.

If a ball is dropped from a specific height onto a fixed surface, then the height of the rebound and the drop height can be used to determine the coefficient of restitution. This is a measure of the bounciness of the ball and the implement or surface it strikes will greatly affect the outcome.



$$e = \sqrt{1.01 / 1.80} = 0.75 \cdot \text{it comes back to } 56\% \text{ of the height}$$

Elastic collisions

An elastic collision occurs when the two objects collide and then bounce off one another.

A collision between two pool or snooker balls is a good example of an almost totally elastic collision. The objects collide and then seem to bounce off of each other and move separately.

In a perfectly elastic collision, in a straight line, each object transfers all of its momentum to the other object.

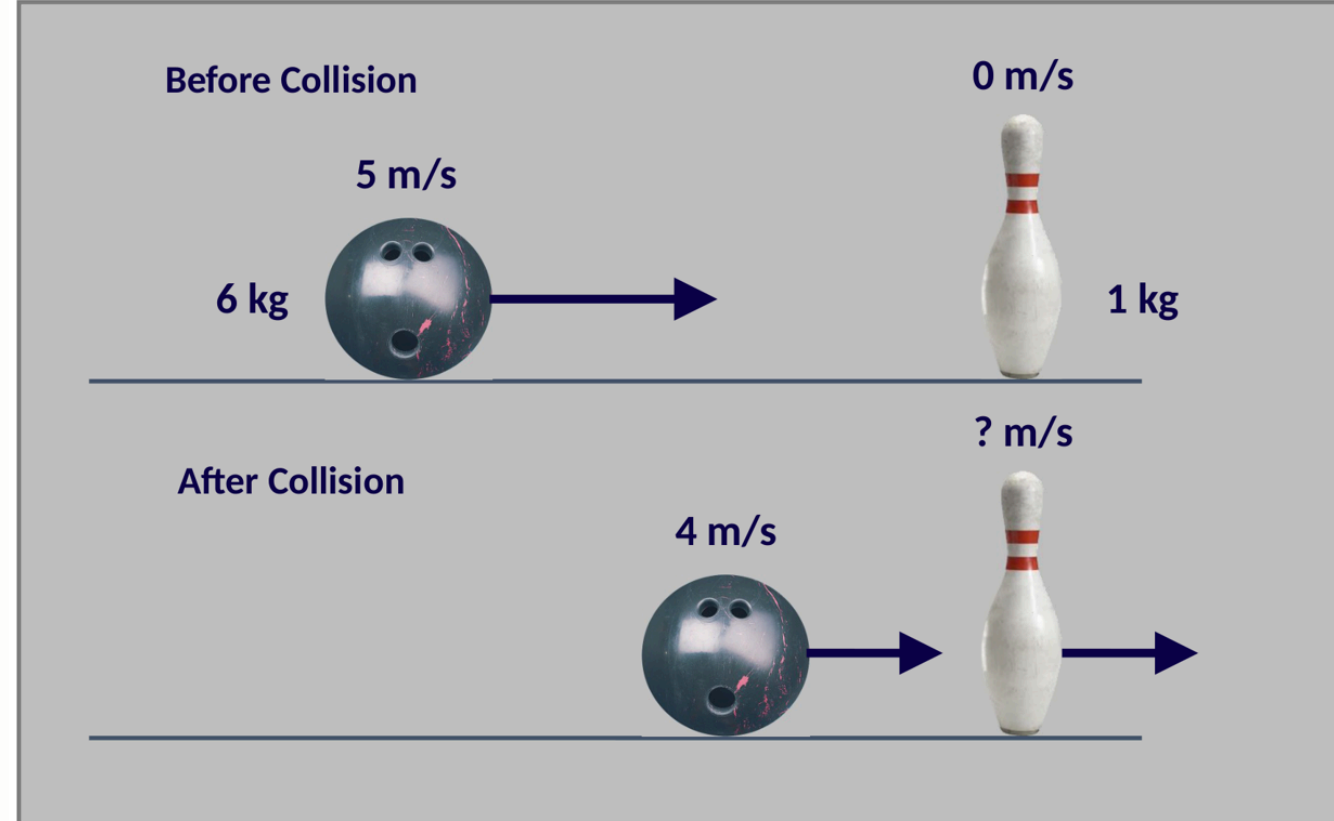
$$m_A v_{A \text{ initial}} + m_B v_{B \text{ initial}} = m_A v_{A \text{ final}} + m_B v_{B \text{ final}}$$



Example Problem

A 6 kg bowling ball enters an elastic collision with a 1 kg bowling pin. If the bowling ball has a velocity of 5 m/s before the collision and 4 m/s after the collision, what is the speed of the pin after the collision? (Assume that the speed of the pin before the collision is 0 m/s)





Solution

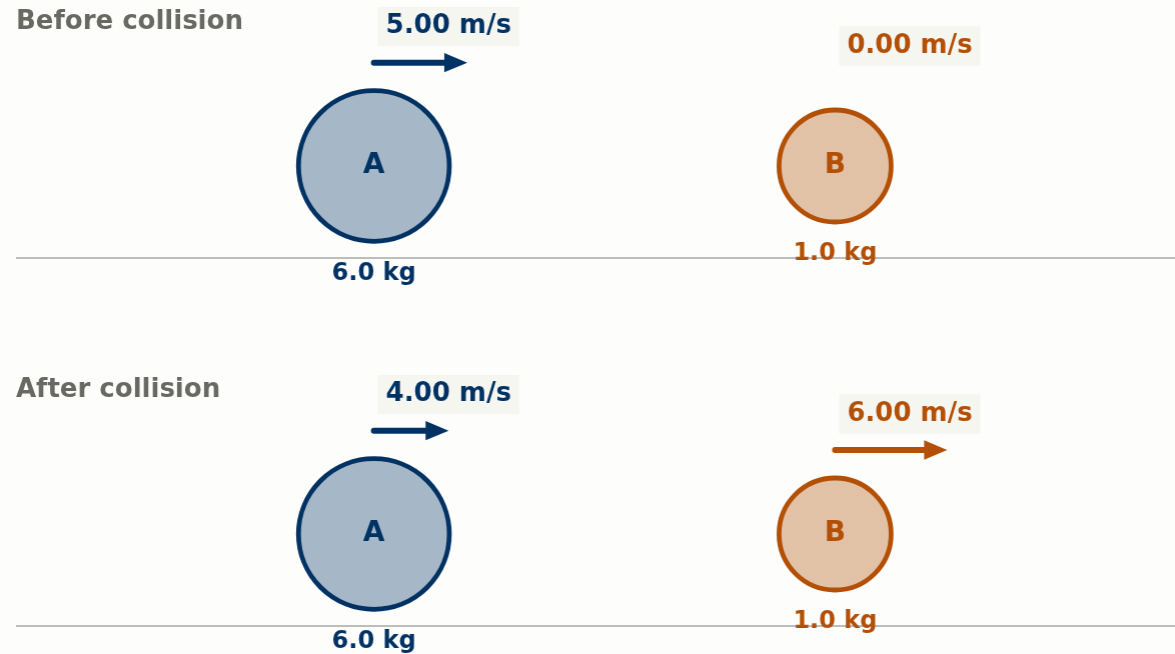
Before collision

After collision

$$\begin{aligned}(m_{\text{ball1}})(v_{\text{ball1}}) + (m_{\text{pin1}})(v_{\text{pin1}}) &= (m_{\text{ball2}})(v_{\text{ball2}}) + (m_{\text{pin2}})(v_{\text{pin2}}) \\ (6 \text{ kg})(5 \text{ m/s}) + (1 \text{ kg})(0 \text{ m/s}) &= (6 \text{ kg})(4 \text{ m/s}) + (1 \text{ kg})(v_{\text{pin2}}) \\ 30 \text{ kg}\cdot\text{m/s} &= 24 \text{ kg}\cdot\text{m/s} + v_{\text{pin2}}(1 \text{ kg}) \\ v_{\text{pin2}} &= \mathbf{6 \text{ m/s}}\end{aligned}$$

After the collision, the pin has a velocity of 6 m/s in the original direction of the bowling ball.

The same collision, with the numbers on sliders



$e = 0.40$ · partly elastic

momentum is conserved

$$m_A v_{Ai} + m_B v_{Bi} = 30.00 \text{ kg}\cdot\text{m/s}$$

$$m_A v_{Af} + m_B v_{Bf} = 30.00 \text{ kg}\cdot\text{m/s}$$

the restitution equation

$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$
$$= 2.00 / 5.00 = 0.40$$

v_{Af}
4.00

v_{Bf}
6.00

KE kept
88%

A: 5.00 → **4.00** m/s · B: 0.00 → **6.00** m/s

Momentum is the same before and after, always — that is the law. What the coefficient of restitution decides is how it is **shared out**. Between the two, some kinetic energy is lost but they still separate.

bowling ball

equal, B at rest

equal, head-on

2 kg meets 1 kg

inelastic

Andre & Hulk

Elastic collisions — equal masses

Since m is the same we can take it out of the equation. Assume $v = 1$ m/s and $v_{B \text{ initial}}$ is zero. With a coefficient of restitution of 1, **A stops dead and B leaves at 1 m/s.**



$e = 1.00$ · perfectly elastic

momentum is conserved

$$m_A v_{Ai} + m_B v_{Bi} = \mathbf{1.00 \text{ kg}\cdot\text{m/s}}$$

$$m_A v_{Af} + m_B v_{Bf} = \mathbf{1.00 \text{ kg}\cdot\text{m/s}}$$

the restitution equation

$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$

$$= 1.00 / 1.00 = \mathbf{1.00}$$

v_{Af}
0.00

v_{Bf}
1.00

KE kept
100%

A: 1.00 → **0.00** m/s · B: 0.00 → **1.00** m/s

Momentum is the same before and after, always — that is the law. What the coefficient of restitution decides is how it is **shared out**. At $e = 1$ no kinetic energy is

bowling ball

equal, B at rest

equal, head-on

Elastic collisions — head on

Now B is coming the other way at 0.5 m/s. Same two equations, same coefficient of restitution — **they swap velocities.**



$e = 1.00$ · perfectly elastic

momentum is conserved

$$m_A v_{Ai} + m_B v_{Bi} = 0.50 \text{ kg}\cdot\text{m/s}$$

$$m_A v_{Af} + m_B v_{Bf} = 0.50 \text{ kg}\cdot\text{m/s}$$

the restitution equation

$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$
$$= 1.50 / 1.50 = 1.00$$

v_{Af}
-0.50

v_{Bf}
1.00

KE kept
100%

A: 1.00 → -0.50 m/s · B: -0.50 → 1.00 m/s

Momentum is the same before and after, always — that is the law. What the coefficient of restitution decides is how it is **shared out**. At $e = 1$ no kinetic energy is lost at all.

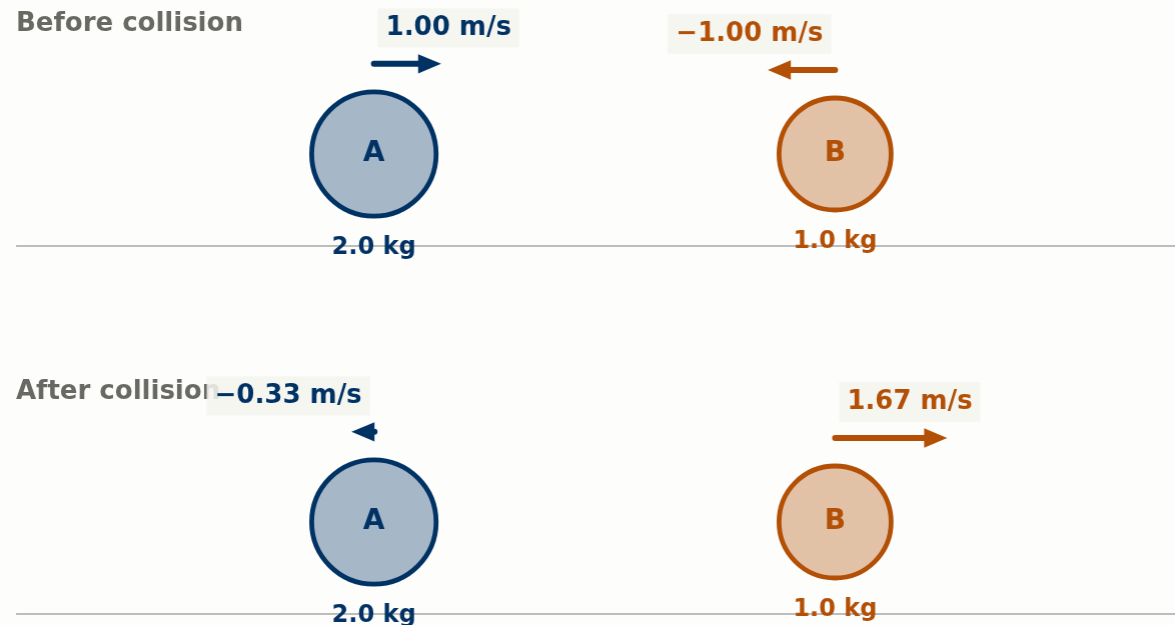
bowling ball

equal, B at rest

equal, head-on

Elastic collisions — different masses

A 2 kg object at 1 m/s meets a 1 kg object at −1 m/s. The heavier one is turned around; the lighter one is thrown back much faster than it arrived.



$e = 1.00$ · perfectly elastic

momentum is conserved

$$m_A v_{Ai} + m_B v_{Bi} = 1.00 \text{ kg}\cdot\text{m/s}$$

$$m_A v_{Af} + m_B v_{Bf} = 1.00 \text{ kg}\cdot\text{m/s}$$

the restitution equation

$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$
$$= 2.00 / 2.00 = 1.00$$

v_{Af}
−0.33

v_{Bf}
1.67

KE kept
100%

A: 1.00 → −0.33 m/s · B: −1.00 → 1.67 m/s

Momentum is the same before and after, always — that is the law. What the coefficient of restitution decides is how it is **shared out**. At $e = 1$ no kinetic energy is lost at all.

bowling ball

equal, B at rest

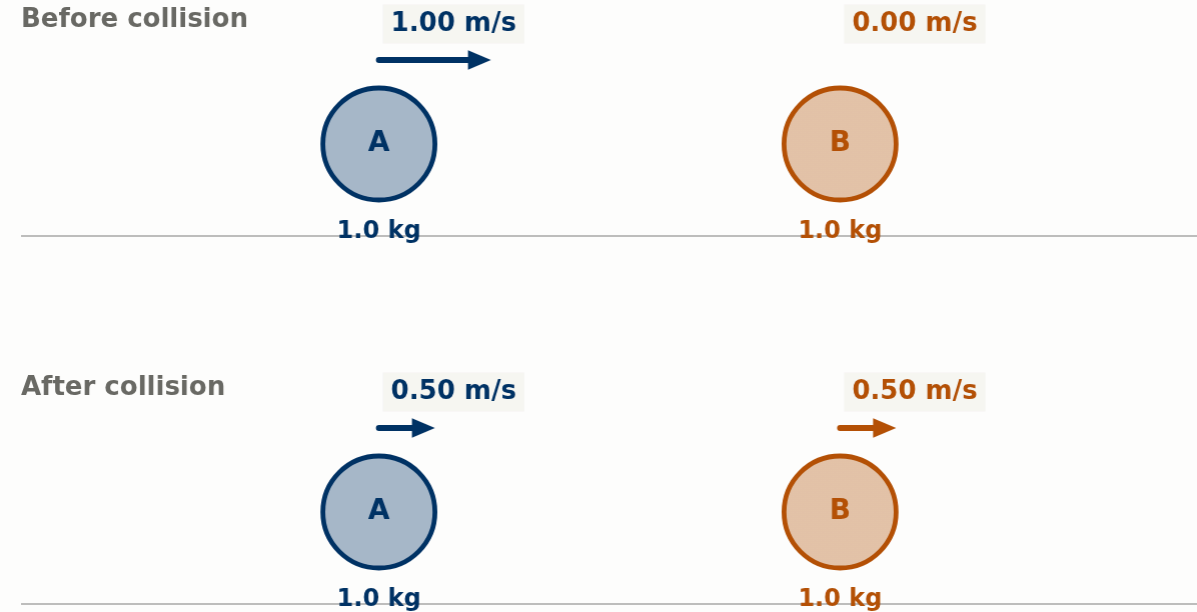
equal, head-on

Inelastic collisions

An inelastic collision occurs when two objects collide and they behave as one object after the collision.

A common example of a perfectly inelastic collision is when two objects collide and then stick together afterwards. This equation describes the conservation of momentum for an inelastic collision:

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_{\text{combined}}$$



$e = 0.00$ · perfectly inelastic

momentum is conserved

$$m_A v_{Ai} + m_B v_{Bi} = \mathbf{1.00} \text{ kg}\cdot\text{m/s}$$

$$m_A v_{Af} + m_B v_{Bf} = \mathbf{1.00} \text{ kg}\cdot\text{m/s}$$

the restitution equation

$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$

v_{Af}
0.50

v_{Bf}
0.50

KE kept
50%

Example Problem

During the feature match of Wrestlemania III, Hulk Hogan and Andre the Giant run towards each other and collide. If Andre has a mass of 235 kg and was traveling 1.4 m/s, and if Hulk has a mass of 115 kg and was traveling at -3.8 m/s, what is their combined velocity? (assuming that they entangle and continue traveling as one)



Solution

- given in this question:

$$m_1 = 235 \text{ kg} \quad m_2 = 115 \text{ kg}$$
$$v_1 = 1.4 \text{ m/s} \quad v_2 = -3.8 \text{ m/s}$$

- Momentum Before Collision = Momentum After Collision

$$(mv)_1 + (mv)_2 = (m_1 + m_2)v$$

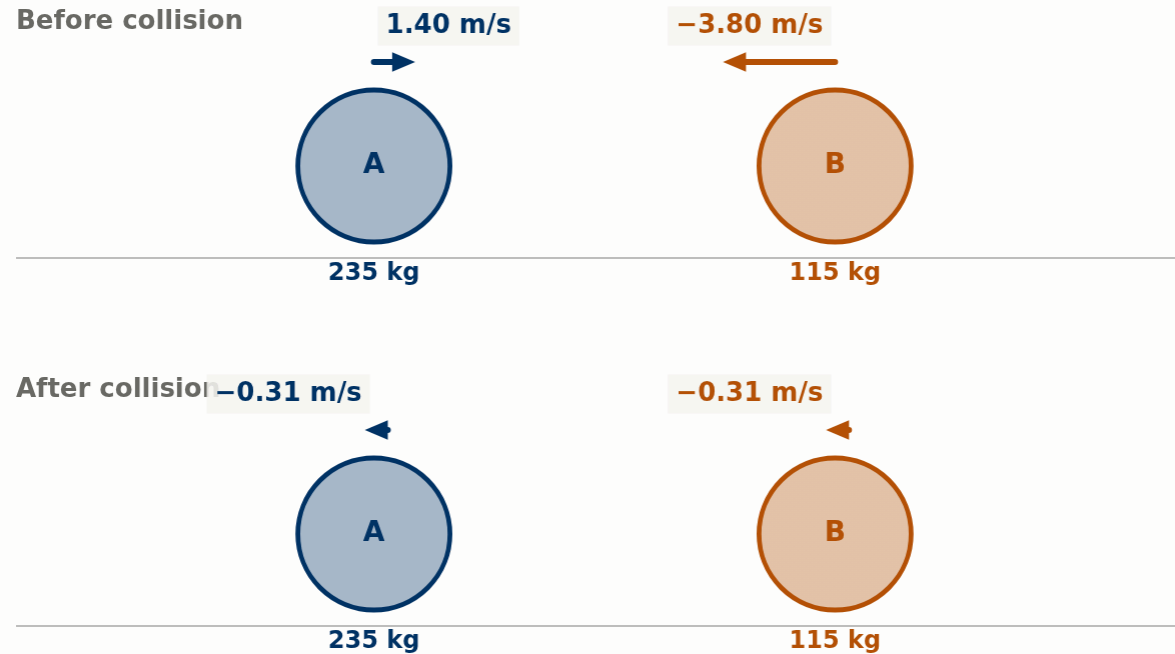
$$(235 \text{ kg})(1.4 \text{ m/s}) + (115 \text{ kg})(-3.8 \text{ m/s}) = (235 \text{ kg} + 115 \text{ kg})v$$

$$329 \text{ kg}\cdot\text{m/s} - 437 \text{ kg}\cdot\text{m/s} = (350 \text{ kg})(v)$$

$$v = \mathbf{-0.3 \text{ m/s}}$$

Their combined velocity is 0.3 m/s in Hulk Hogan's original direction

The same collision, with the numbers on sliders



$e = 0.00$ · perfectly inelastic

momentum is conserved

$$m_A v_{Ai} + m_B v_{Bi} = -108.00 \text{ kg}\cdot\text{m/s}$$

$$m_A v_{Af} + m_B v_{Bf} = -108.00 \text{ kg}\cdot\text{m/s}$$

the restitution equation

$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$
$$= 0.00 / 5.20 = 0.00$$

v_{Af}
-0.31

v_{Bf}
-0.31

KE kept
2%

A: 1.40 → -0.31 m/s · B: -3.80 → -0.31 m/s

Momentum is the same before and after, always — that is the law. What the coefficient of restitution decides is how it is **shared out**. At $e = 0$ they leave together at one velocity and the lost kinetic energy has gone into deformation and heat.

bowling ball

equal, B at rest

equal, head-on

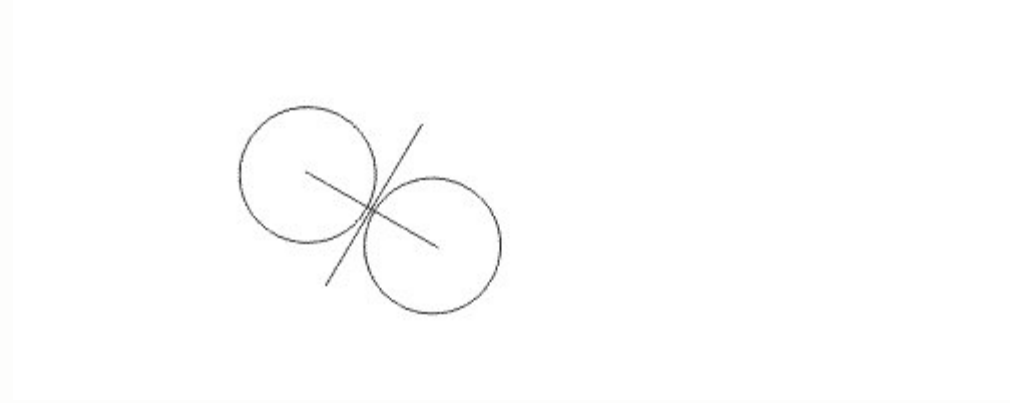
2 kg meets 1 kg

inelastic

Andre & Hulk

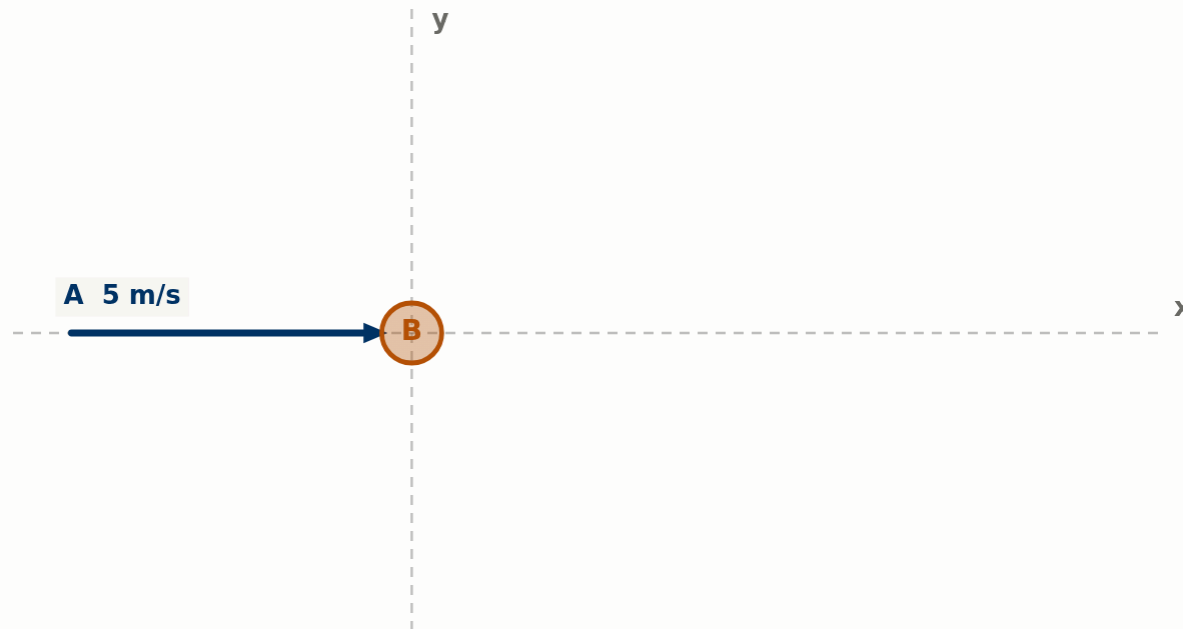
You are required to be able to solve questions for conservation of momentum only on a **straight line** — 1 dimension.

However you will be expected to **conceptually understand** simple collisions in 2 dimensions.



Collisions in two dimensions

For elastic collisions in two dimensions we need to break the momentum into components. Each ball has a mass of 0.1 kg and ball B was at rest. What is the speed and direction of ball B after the collision?



the set-up

Each ball has a mass of 0.1 kg and ball B was at rest. What is the speed and direction of ball B after the collision?

$$P_o = P^f$$

$$P_{ox} = P_x^f, P_{oy} = P_y^f$$

B leaves at **3.42 m/s**, 17.0° below the x axis

The y momentum started at zero, so whatever A takes upward, B must take downward by exactly as much. Two equations, one for each axis — that is the whole of two-dimensional conservation.

1 · the set-up

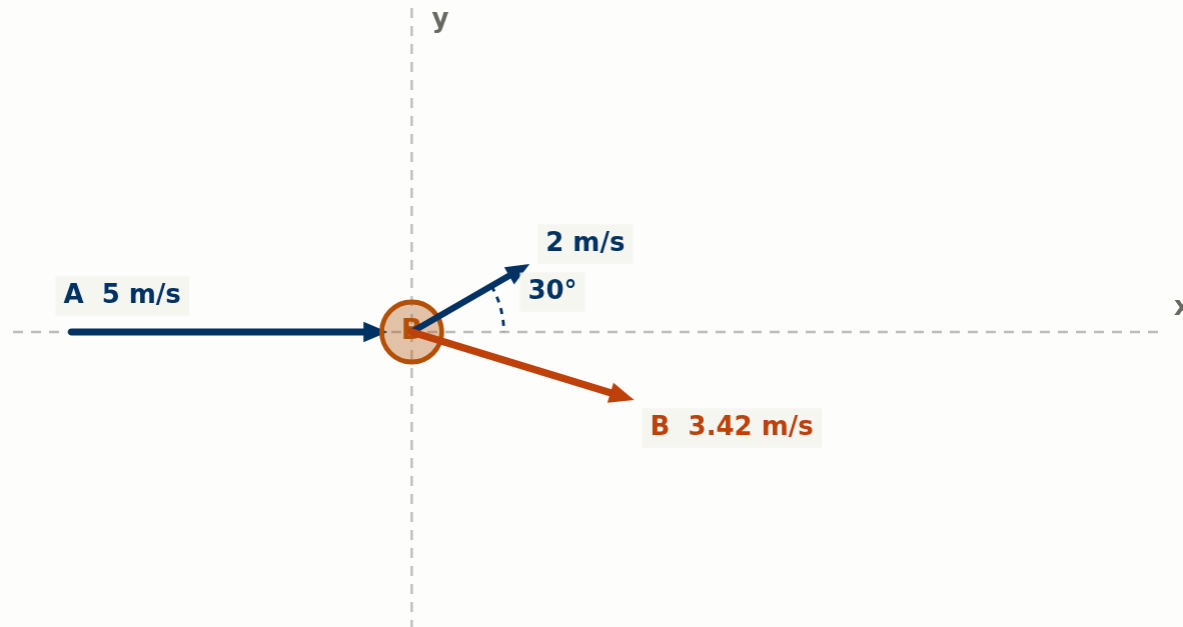
2 · A leaves

3 · components

4 · the answer

Collisions in two dimensions

$P_{ox} = P_{fx}$ and $P_{oy} = P_{fy}$ — one equation for each axis.



momentum in x and y

$$\begin{aligned}P_{ox} &= P_{fx} \\(0.1)(5) + 0 &= (0.1)(2 \cos 30) + (0.1)V_{B_x}^f \\V_{B_x}^f &= \mathbf{3.27 \text{ m/s}} \\P_{oy} &= P_{fy} \\0 &= (0.1)(2 \sin 30) + (0.1)V_{B_y}^f \\V_{B_y}^f &= \mathbf{-1.00 \text{ m/s}}\end{aligned}$$

B leaves at **3.42 m/s**, 17.0° below the x axis

The y momentum started at zero, so whatever A takes upward, B must take downward by exactly as much. Two equations, one for each axis — that is the whole of two-dimensional conservation.

A before  **5.0 m/s**

1 · the set-up

2 · A leaves

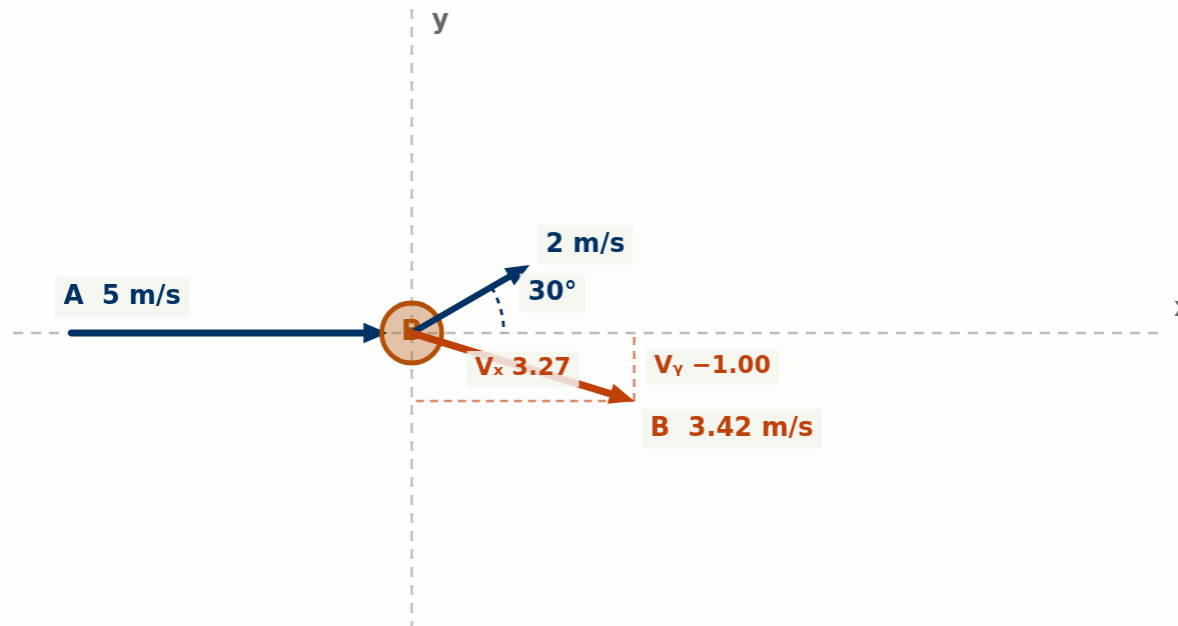
3 · **components**

4 · the answer

A after  **2.0 m/s**

Collisions in two dimensions

Put the two components back together for the magnitude and the direction.



magnitude and direction

$$V_B^{f2} = V_x^2 + V_y^2$$

$$V_B^f = \mathbf{3.42} \text{ m/s}$$

$$\tan\theta = V_y / V_x$$

$$\theta = \mathbf{17.01^\circ} \text{ below the x axis}$$

B leaves at **3.42 m/s**, 17.0° below the x axis

The y momentum started at zero, so whatever A takes upward, B must take downward by exactly as much. Two equations, one for each axis — that is the whole of two-dimensional conservation.

1 · the set-up

2 · A leaves

3 · components

4 · the answer

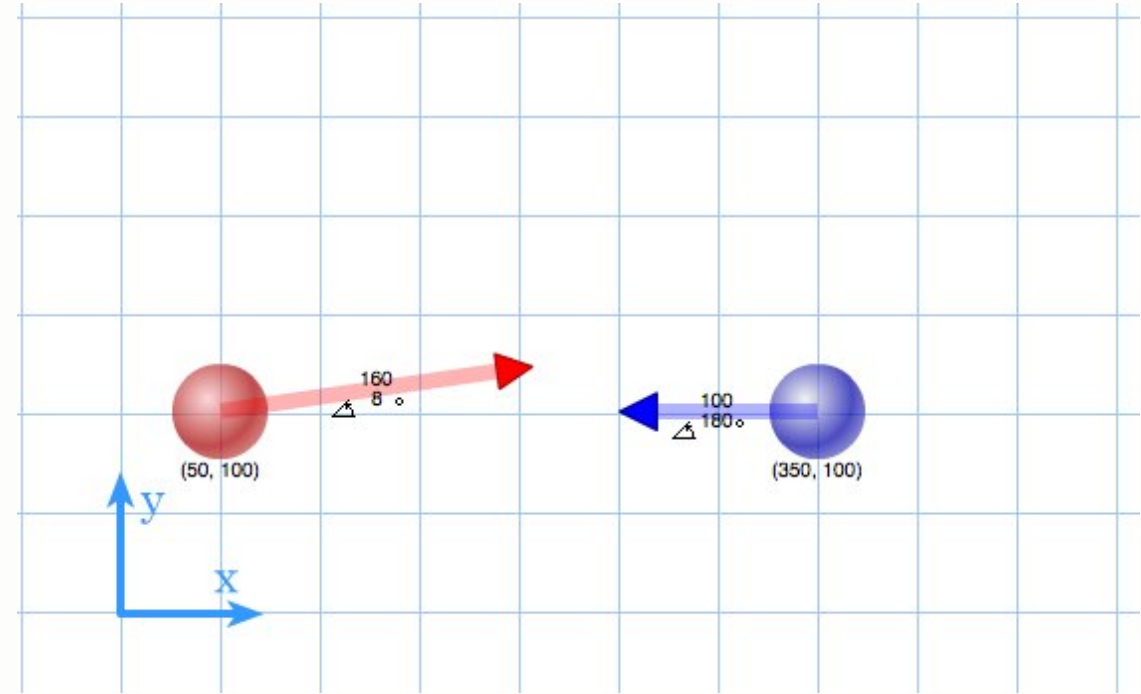
A before  5.0
m/s

A after  2.0
m/s

Linear Kinetics

In the previous example we had an idea of the angle of the balls velocities before and after the collision. This allowed us to determine the magnitude of the velocity. However that is not always the case.

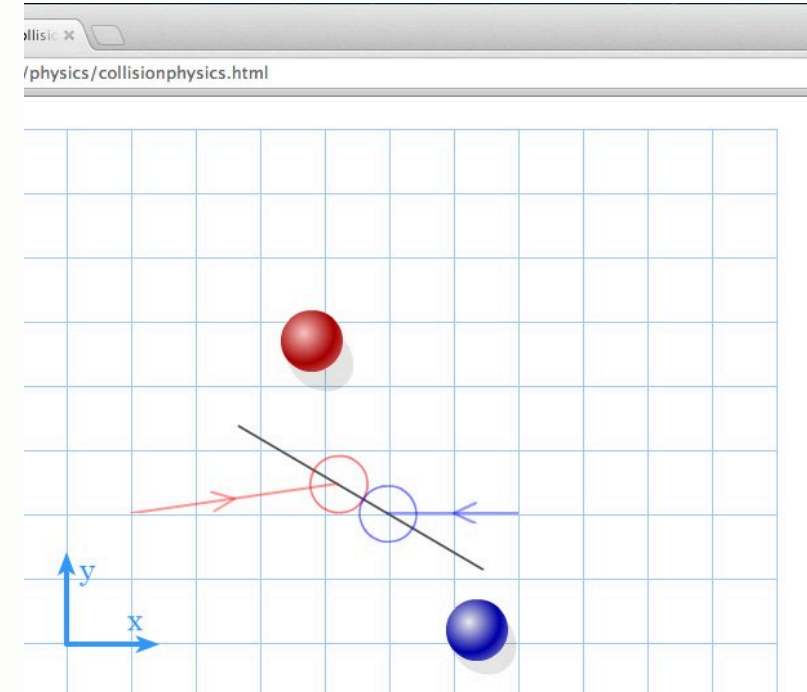
To determine the angle and magnitude of the balls velocities before and after the collision we need to set up a frame of reference and explore the conservation of momentum in axes **normal** (perpendicular) and **tangent** (parallel) to the contact surface interaction.



Normal and tangent axes

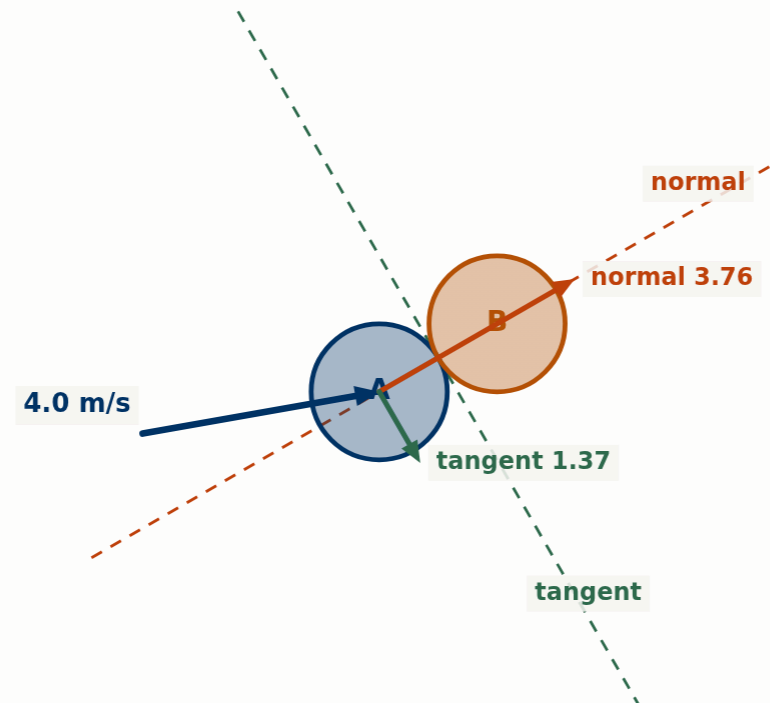
With our original x and y axis you can see:

- the velocities of the balls before the collision
- The axis **normal** (perpendicular) to the contacting surfaces
- The axis **tangent** (parallel) to the contacting surfaces



Normal and tangent components

To determine the final velocity vectors angles you need to set up a normal and tangent coordinate system. **The tangential components remain unchanged**; conservation of linear momentum is maintained for the **normal** components.

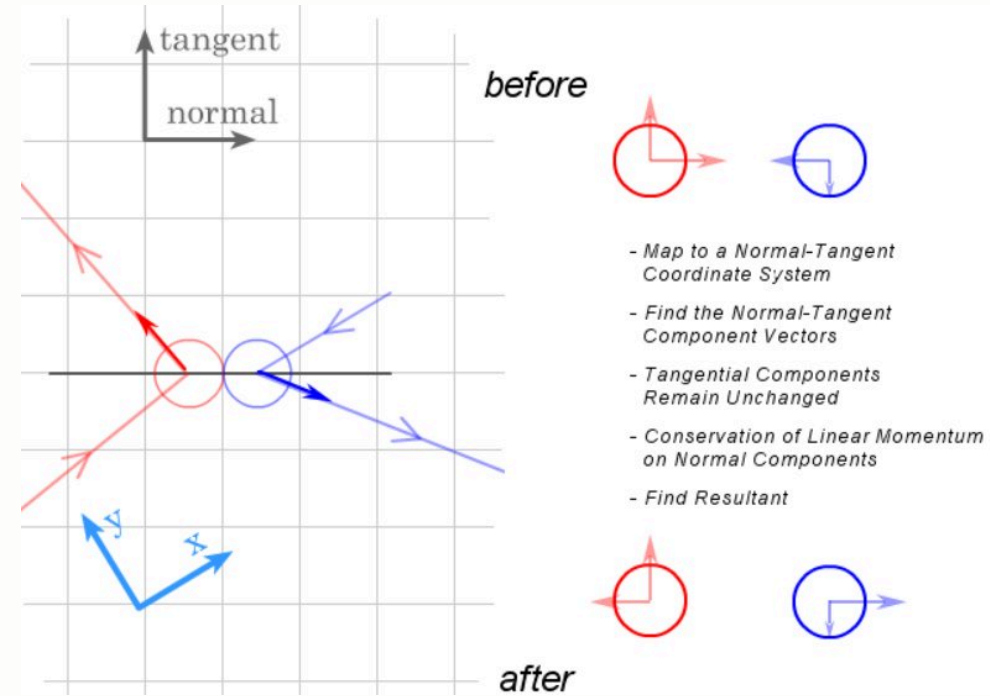


before the collision · normal **3.76** m/s · tangent **1.37** m/s

Linear Kinetics

Let's think about that in terms of Newton's 1st law.

An object at rest tends to stay at rest and an object in motion tends to stay in motion (**in its component direction**) unless a net external force acts upon it.



Transfer of momentum

Understanding Momentum is really important when dealing with the combined motion of the human body and techniques to **transfer momentum from one limb to another.**

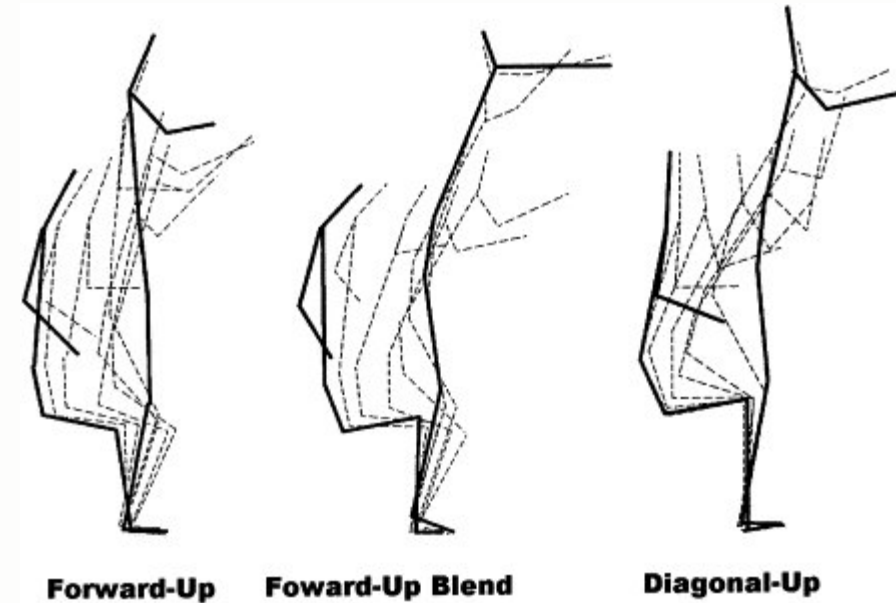
The outcome of transferring momentum from one limb to another can be explained by the conservation of momentum of the system.



Transfer of momentum in sit to stand

The transfer of momentum during human movement can be discussed in reference to simple and complex tasks.

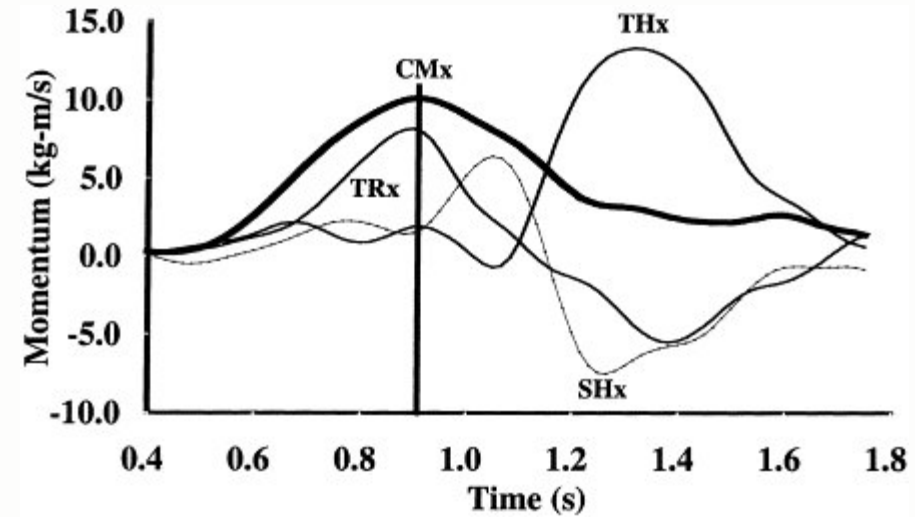
In a sit to stand task we use transfer of momentum to help us stand. By leaning forward our trunk has momentum which can be used to assist in hip and leg extension.



Horizontal momentum during sit to stand

Dark vertical line indicates time of occurrence of peak horizontal momentum of the CM.

- CMx is the horizontal momentum of whole body's CM
- TRx is the horizontal momentum of the trunk
- THx is the horizontal momentum of the thigh
- SHx is the horizontal momentum of the shank

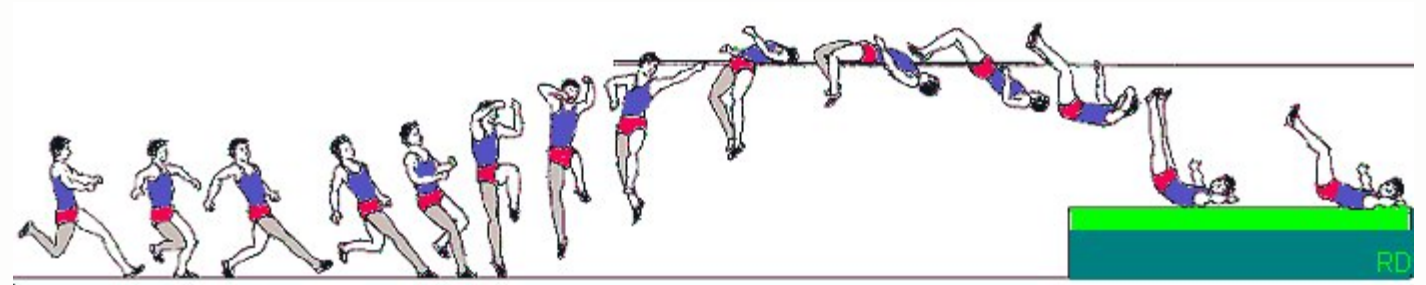


Transfer of momentum in the high jump

During a complex task such as high jump we use transfer of momentum to help us increase our **vertical velocity**.

The free limbs contribute by transferring their individual momentum to the velocity of the whole body:

- Contralateral arm relative momentum
- Ipsilateral arm relative momentum
- Lead leg relative momentum
- Combined free limb relative momentum



$$\text{CARM} = m_{\text{CA}} V_{\text{CA/CS}}$$

$$\text{IARM} = m_{\text{IA}} V_{\text{IA/IS}}$$

$$\text{LLRM} = m_{\text{LL}} V_{\text{LL/H}}$$

Free limb relative momentum

- This shows that the RM of the arms are negative at touch-down as the arms are being driven downwards
- the leg is flexing at the knee giving its CM a net upward velocity
- The RM of the leg continues to increase as it is flexed and elevated
- The RM of arms becomes positive as they reach the bottom of their swing and are elevated
- The RM of each arm are similar and they reach their peak at approximately the same point in time but that of the lead leg reaches its peak earlier than the arms

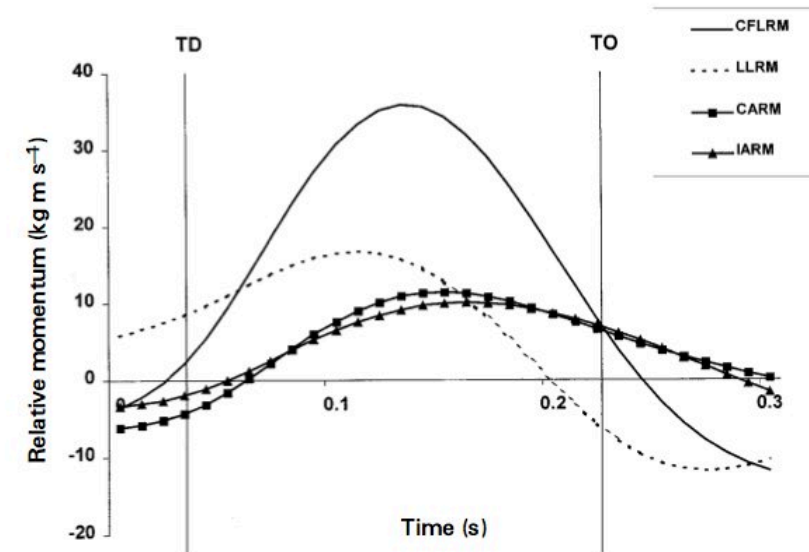


Figure 1. Vertical relative momentum for individual free limbs (CARM= contralateral arm relative momentum, IARM= ipsilateral arm relative momentum, LLRM= lead leg relative momentum) and combined free limbs relative momentum (CFLRM) for athlete Austin.

Free limb relative momentum

The combined free limb RM (CFLRM) reaches a peak just after the lead leg but just before the arms. The relative momentum of all free limbs **reduces at take-off**, illustrating the transfer of each segment momentum to the rest of the body.

The graph shows the estimated additional vertical force generated as a result of the transfer of momentum.

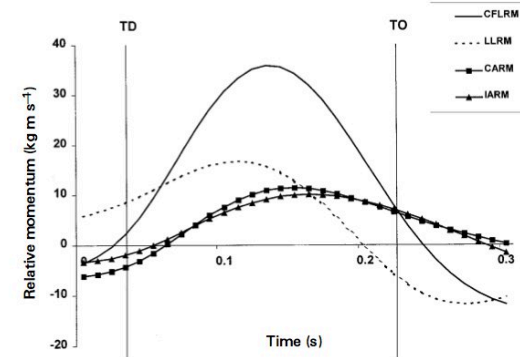


Figure 1. Vertical relative momentum for individual free limbs (CARM= contralateral arm relative momentum, IARM= ipsilateral arm relative momentum, LLRM= lead leg relative momentum) and combined free limbs relative momentum (CFLRM) for athlete Austin.

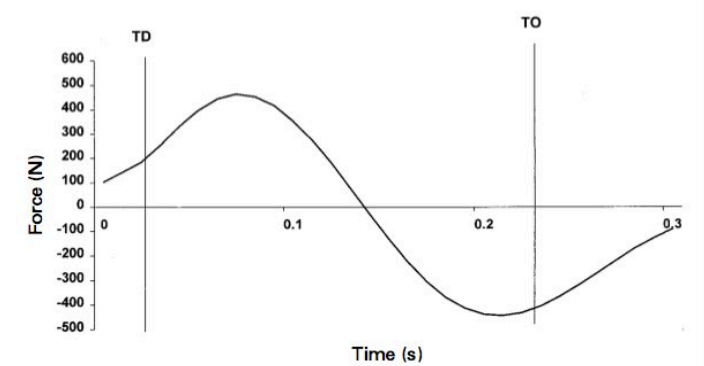


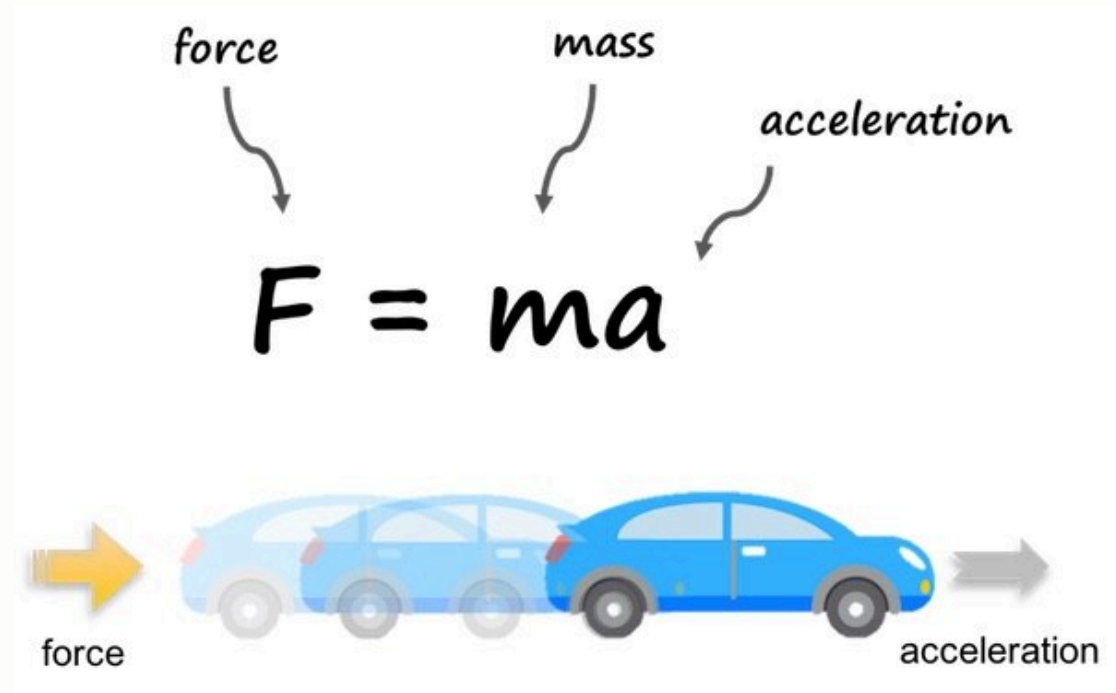
Figure 4. Typical graph of the estimated additional force generated as a result of limb motion.

2nd Law: The Law of Acceleration

Law II

“A force applied to a body causes an acceleration of that body of a magnitude proportional to the force, in the direction of the force, and inversely proportional to the body’s mass.”

$$F = ma$$



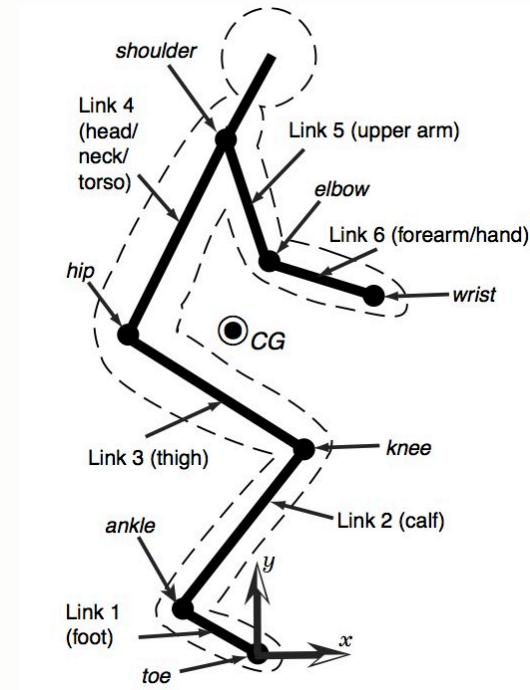
2nd Law: The Law of Acceleration

Newton's second law gives us the connection between **force (F)** and **motion (a)**.

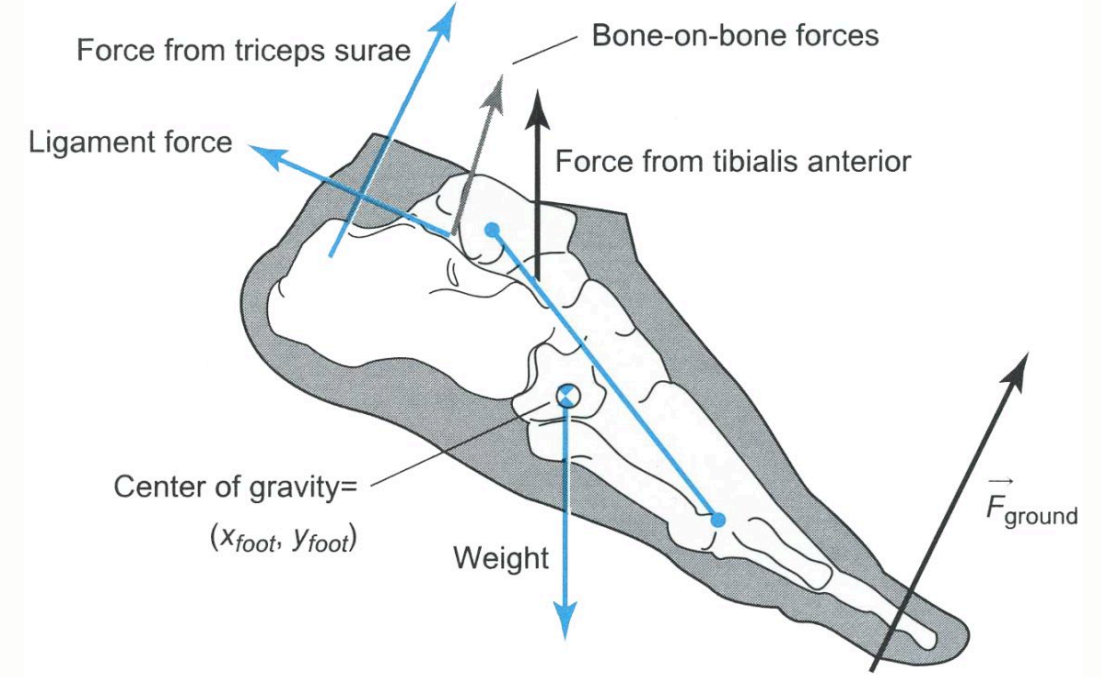
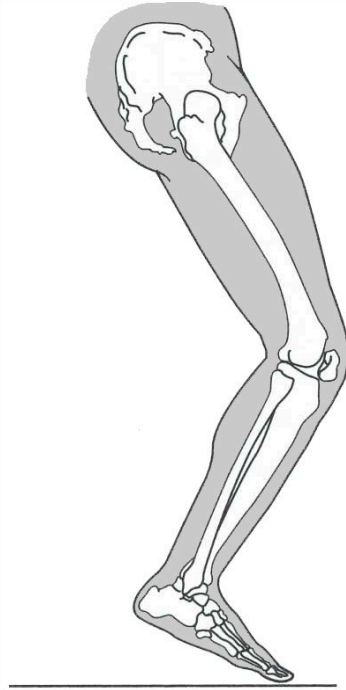
It provides a framework for thinking about how all of the forces (internal and external) can control motion.

We will use this law as a basis for our discussion of static and dynamic analysis including **inverse dynamics**.

$$F = ma$$



Free Body Diagram



3rd Law: The Law of Reaction

Law III

“Every applied force is accompanied by a reaction force, equal in magnitude and opposite in direction.”

- each force of the pair (the action force and the reaction force) is applied to a **different body**
- for example, when a jumper generates a force against the ground, the ground generates an equal and opposite force against the jumper, causing the jumper to accelerate upward

$$\begin{aligned}\Sigma F &= N - W \\ &= 687 - 687 = 0 \text{ N} \\ a &= 0.00 \text{ m/s}^2\end{aligned}$$



ground reaction **1.00 × body weight** · net **0 N** · **a = 0.00 m/s²**

The two big arrows are the **action and reaction pair**, and they are always equal and opposite — but they act on **different bodies**, so they never cancel each other. Only the forces on the jumper decide what the jumper does, and there the ground reaction is up against weight. Right now they balance exactly and nothing accelerates.

Body mass ————— **70 kg**

3rd Law: The Law of Reaction



When a jumper generates a force against the ground, the ground generates an equal and opposite force against the jumper.